M339D: March 25th, 2026.

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Problem Set #10

Binomial option pricing: Two or more periods.

Problem 10.1. For a two-period binomial model, you are given that:

(1) each period is one year; $h=1$
(2) the current price of a non-dividend-paying stock S is $S(0) = \$20$;
(3) $u = 1.2$, with u as in the standard notation for the binomial model;
(4) $d = 0.8$, with d as in the standard notation for the binomial model;
(5) the continuously compounded risk-free interest rate is $\boxed{r = 0.04}$

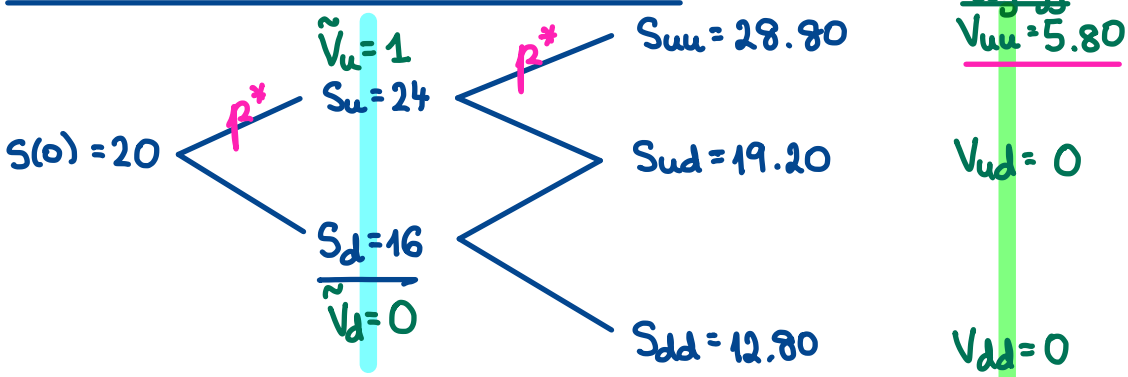
Consider a **special** call option which pays the excess above the strike price $K = 23$ (if any!) at the end of **every** binomial period.

Find the price of this option.

→: 1st ✓ Tree ✓ + Payoff ✓

2nd $p^* = ?$ ✓

3rd Risk-Neutral Pricing Formula



$$p^* = \frac{e^{rh} - d}{u - d} = \frac{e^{0.04} - 0.8}{1.2 - 0.8} = \underline{0.602}$$

$$\tilde{V}(0) = e^{-0.04} \cdot p^* \cdot 1 = \underline{0.5784}$$

$$V(0) = e^{-0.04(2)} \cdot (p^*)^2 \cdot 5.80 = \underline{1.9413}$$

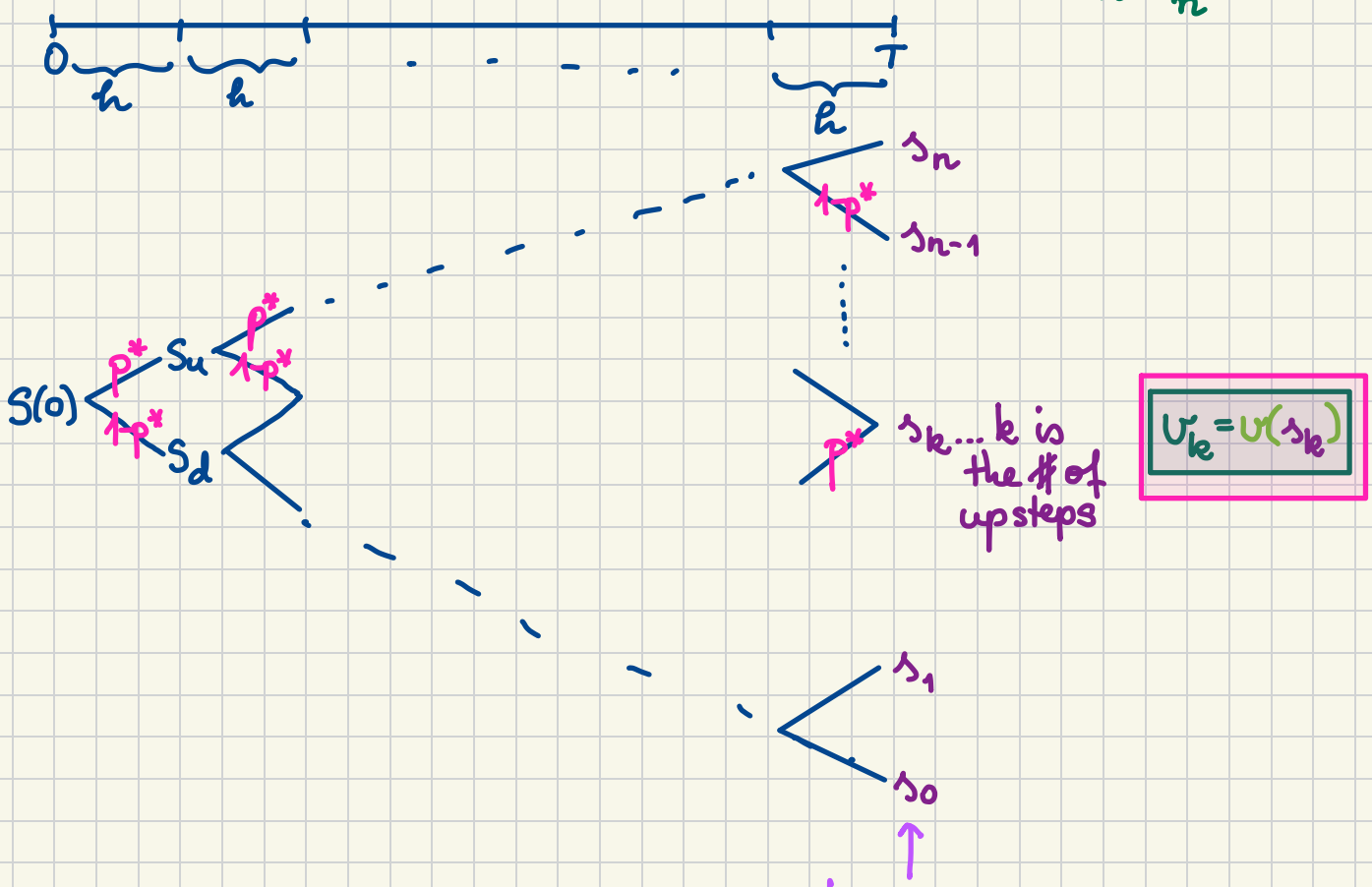
+

answer: price of the special call = 2.5197 □

Multiple Binomial Periods.

T ... exercise date of a European option
n ... # of periods

} the length of each period: $h = \frac{T}{n}$



The $n+1$ values are our support of $S(T)$.

$\Rightarrow$ for every $k = 0, 1, \dots, n$:

$$s_k = S(0) \cdot u^k \cdot d^{n-k} = S(0) \left(\frac{u}{d}\right)^k d^n$$

k .... # of upsteps

Consider a European option w/ payoff f'tion $v(\cdot)$.
Then, the possible values of the payoff are

$$v_k = v(s_k) \quad \text{for } k = 0, 1, \dots, n$$

Risk-Neutral Pricing.

$$V(0) = e^{-rT} \mathbb{E}^*[V(T)]$$

w/ $p^* = \frac{e^{rh} - d}{u - d}$

$\Rightarrow$ The risk-neutral probability of reaching the payoff v_k is:

$$\boxed{\binom{n}{k}(p^*)^k(1-p^*)^{n-k}} \quad \text{for } k=0,1,\dots,n.$$

$\Rightarrow$ The risk-neutral option price:

$$\boxed{V(0) = e^{-rT}\sum_{k=0}^{n}\left(\binom{n}{k}(p^*)^k(1-p^*)^{n-k}\cdot v_k\right)}$$

Problem 10.2. Let the continuously compounded risk-free interest rate be 0.10. Let the initial price of a non-dividend-paying stock be \$100 per share. You use a five-period binomial tree to model the stock price over the next year. Let $u = 1.04$ and $d = 0.96$.

What is the price of a one-year, at-the-money European call option on the above stock?

$K=100$

→: The risk-neutral probability:

$$p^* = \frac{e^{rh} - d}{u-d} = \frac{e^{0.1\cdot 0.2} - 0.96}{1.04 - 0.96} = \underline{0.7525}$$

The relevant final stock prices in our tree:

$s_5 = S(0)\cdot u^5 = 100(1.04)^5 = \underline{121.67}$ $\quad v_5 = 21.67$

$s_4 = S(0)\cdot u^4 \cdot d = 100(1.04)^4 \cdot (0.96) = \underline{112\ 31}$ $\quad v_4 = 12.31$

$s_3 = S(0)\cdot u^3 \cdot d^2 = 100(1.04)^3 \cdot (0.96)^2 = \underline{103.67}$ $\quad v_3 = 3.67$

The remaining terminal nodes are out-o-money

$\Rightarrow$

$$V(0) = e^{-0.1}\left(21.67\cdot(p^*)^5 + 12.31\cdot\binom{5}{1}(p^*)^4(1-p^*) + 3.67\binom{5}{2}(p^*)^3(1-p^*)^2\right) = \underline{10.01821}$$

$\frac{5\cdot 4}{2} = 10$

$\square$